

HARSHIT RAI

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PROFESSIONAL SUMMARY

AI/ML Engineer with end-to-end expertise in Generative AI, model development, NLP, and production MLOps. Shipped 10+ systems combining fine-tuned LLMs, multi-agent RAG pipelines, vector databases (FAISS), and hardened Kubernetes deployments. Skilled in prompt engineering, model deployment, and CI/CD pipelines. Key results: 96% RAG test pass rate, 60% hallucination reduction, 45% ML pipeline speedup, 90% Docker image size reduction, and sub-200ms response times.

TECHNICAL SKILLS

AI/ML & Generative AI: LangChain, LangGraph, FAISS (Vector Database), RAG Pipelines, Multi-Agent Systems, HuggingFace, BERT, mDeBERTa, TensorFlow, PyTorch, LoRA Fine-Tuning, Prompt Engineering, NLP, Computer Vision
MLOps & Model Deployment: MLflow, Prefect, Airflow, Model Pruning & Quantization, Experiment Tracking, Data Pipeline Optimization, AWS SageMaker
Cloud & Infrastructure: AWS (EC2, S3, EKS, ECR), Docker, Kubernetes (EKS/K3s/Minikube), Helm, HPA, Terraform, Linux
CI/CD & GitOps: Jenkins, ArgoCD, GitHub Actions, DevSecOps, Bandit, Trivy, Cosign, Dockle
Backend: Python, FastAPI, Flask, PostgreSQL, MongoDB, Redis, REST APIs
Monitoring: Prometheus, Grafana, Health/Readiness Probes, Distributed Tracing

EXPERIENCE

Machine Learning Engineer (Intern) | Brainwave Matrix Solutions, Noida Jul 2025 – Sep 2025

- Developed and deployed end-to-end ML models for data pipeline processing using Python and TensorFlow; achieved ~45% performance gain via model pruning and quantization techniques.
- Built scalable Docker + Kubernetes deployment pipelines with automated GitHub Actions CI/CD for model testing, versioning, and release; reduced deployment time by 60%.
- Implemented model monitoring using Prometheus and Grafana; maintained 99.9% uptime across deployed services.

PROJECTS

IntelliAssist — RAG-Enhanced Fine-Tuned GPT-2 Assistant Oct – Nov 2025

- Fine-tuned DistilGPT-2 with LoRA adapters reducing trainable parameters from 124M to ~1M (95% reduction); integrated FAISS vector database + Sentence-Transformers for RAG grounding.
- Achieved 43% faster inference, 70% less VRAM, and ~60% hallucination reduction via prompt engineering and retrieval grounding; domain accuracy improved from 35–40% to 65–75% — no GPU required.
- Added Whisper voice input and Edge TTS output for end-to-end conversational AI pipeline deployable on CPU.

PaperIntel AI — Multi-Agent Research Paper Analysis Jan – Feb 2026

- Designed 10-agent LangGraph pipeline (router → retriever → summarizer → flaw detector → PPT generator) processing PDFs/arXiv IDs into structured summaries in under 30 seconds using SciBERT + FAISS + Groq llama-3.3-70b.
- Deployed via ArgoCD GitOps, Jenkins CI/CD, Kubernetes with GROBID for PDF parsing, Redis caching, and S3 vector storage; achieved 96% test pass rate across pipeline agents.

HireSparkAI — AI-Powered Talent Intelligence Platform Feb – Mar 2026

- Built dual-pipeline RAG platform for bulk resume parsing, NLP-based skill extraction, and FAISS vector search candidate routing with ranked, explainable match scores.
- Delivered AI resume optimisation, interview Q-gen, and job-fit prediction; deployed on K3s with ArgoCD, Jenkins CI/CD, Prefect data pipelines, HPA auto-scaling, Prometheus observability, and AWS S3 backup.

Secure MLOps Recommendation Engine — NCF + DevSecOps Dec 2025 – Jan 2026

- Built Neural Collaborative Filtering model on MovieLens 100k in a hardened FastAPI service with Jenkins CI/CD pipeline running Bandit (SAST), Trivy, Dockle, and Cosign image signing.
- Implemented Prometheus metrics on latency and CTR, MLflow experiment tracking, Redis caching, and Locust load testing; deployed via ArgoCD with pod disruption budgets and resource quotas.

Indic NLP — Multilingual Sentiment & Topic Classifier Mar – Apr 2026

- Built production FastAPI microservice combining multilingual BERT (5-star sentiment) + mDeBERTa (zero-shot NLP, 100+ languages) for Hinglish/Hindi/English code-mixed text.
- Achieved sub-second CPU batch inference on 20 texts; full MLOps layer with Redis async logging, SlowAPI rate limiting, Kubernetes HPA, Jenkins CI/CD, and health/readiness probes.

RESEARCH

Early Detection of Epilepsy Using Neuroimaging | Sharda University Ongoing

- Deep learning pipeline for early epilepsy detection using EEG/MRI data; targeting pre-ictal patterns using computer vision and signal processing techniques for clinical AI intervention.
- Focus on model interpretability (Grad-CAM, SHAP) to ensure clinical trustworthiness and compliance with medical AI standards.

CERTIFICATIONS

MLOps Fundamentals (Google, Apr 2026) • **MLOps for Generative AI** (Google, Apr 2026) • **Prompt Design in Vertex AI** (Google, Nov 2024) • **AWS SageMaker ML Projects** (Udemy, Apr 2026) • **AWS Cloud Practitioner CLF-C02** (Udemy, Apr 2026) • **Decoding DevOps with AI** (Udemy, Apr 2026)

EDUCATION

B.Tech, Computer Science | Sharda University, Greater Noida, India Aug 2023 – May 2027
Coursework: Machine Learning, Systems Design, Data Structures, Cloud Computing, NLP, Computer Vision